Electrical Power and Energy Systems 160 (2024) 110076

Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

# Hierarchical robust Day-Ahead VPP and DSO coordination based on local market to enhance distribution network voltage stability

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A R T I C L E I N F O

*Keywords:*
Virtual Power Plant
Local Energy Market
Local Flexibility Market
VPP-DSO Coordination
Robust Optimization

A B S T R A C T

As distributed energy resources (DER) are increasing, a variety of business models and markets are appearing. Business models such as virtual power plants (VPPs) contribute to enhanced stability of the power system while improving the profits of participating resources. However, the uncertainty of the behavior of DERs and the resulting stability issues are a major barrier to DER deployment and require a coordination framework between grid operators and business models to overcome. Therefore, in this paper, a hierarchical robust day-ahead co-ordination of a VPP and distribution system operator (DSO) based on the local market is proposed to improve the voltage stability of the distribution network by considering the uncertainty of DERs. For fairness reasons, the contribution of DERs to the distribution system can be addressed by a market-based approach. Local markets are divided into local energy markets (LEM) and local flexibility markets (LFM), which can consider the uncertainty and profitability of DERs. First, the DER determines the seller and buyer status of the LEM based on its self-scheduling with expected prices and is matched based on a double-auction mechanism. At this time, the worst-case scenario of load and renewable energy generation predicted by the Gaussian process (GP) is considered. The mismatched amounts in the LEM are aggregated by the VPP and bid into the wholesale energy market (WEM) operated by the TSO (Transmission system operator). Subsequently, DSO operates a LFM that coordinates microgrid (MG) and VPP on the distribution grid to increase voltage stability. The LFM and WEM problems are considered as interval optimization in the form of mixed integer linear programming and are applied together. The flexibility supply resulting from the adjustment of VPP is compensated by a loss-sensitivity-based flexibility price. Since voltage stability exists in upper and lower bounds, the upper and lower bound of net load through GP-based forecasting is considered along with the linearized network constraints. The results are validated through Monte-Carlo simulation-based scenarios on IEEE benchmark system in MATLAB environment. As a result, it is proved that the proposed model improves the distribution network voltage stability and the profit of the participants.

## 1. Introduction

Recently, the need for energy transition through the expansion of distribution energy resource (DER) to reduce carbon due to the climate crisis has been increasing [1]. With this change, the power system is undergoing a paradigm change from a one-directional system that supplies power from large power plants to a bi-directional system that also generates power from the demand side. However, the increase of DERs can cause various problems in the power system, such as voltage instability due to reverse power flow and demand imbalance due to difficulty in forecasting [2].

On the other hand, several business models based on DER are appearing, including peer-to-peer (P2P) energy trading, aggregators, and virtual power plants (VPPs) [3]. P2P energy trading provides direct energy trading between energy prosumers and consumers with small-scale solar and energy storage systems (ESS) to achieve profits. A model that aggregates DERs to participate in a larger market is an aggregator. A more scaled application is the VPP. VPPs aggregate various types of DERs to participate in markets or provide reserve or flexibility to the power system. In other words, a VPP is a sharing economy-based business model that aggregates DERs and operates them as a single power plant. The ability to aggregate and operate DERs with uncertainty provides a solution to many of the issues raised by DERs. For example, [4–7] proposed a method to improve profits by participating in the day-ahead energy and reserve power markets under the condition of

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https://doi.org/10.1016/j.ijepes.2024.110076
Received 3 November 2023; Received in revised form 24 April 2024; Accepted 4 June 2024
Available online 20 June 2024

## Nomenclature

*Abbreviations*

| | |
|---|---|
| DER | Distributed energy resources |
| LEM | Local energy market |
| WEM | Wholesale energy market |
| VPP | Virtual power plant |
| DSO | Distribution system operator |
| TSO | Transmission system operator |
| GP | Gaussian process |
| ESS | Energy storage system |
| P2P | Peer to peer |
| MILP | Mixed integer linear programming |
| OPF | Optimal power flow |
| MCS | Monte-Carlo simulation |
| SOC | State of charge |
| LSF | Load shift factor |
| GL | General Load |
| MG | Microgrid |
| pf | Power factor |
| SS | Self - Scheduling |

*Decision variables*

| | |
|---|---|
| $P^{t,i}_{exp,sell}/P^{t,i}_{exp,buy}$ | Expected sell/buy [MW] of $i^{th}$ node at time t |
| $P^{t,i}_{SC,ch}/P^{t,i}_{SC,dch}$ | Charge/Discharge [MW] of $i^{th}$ participant at time t for SS |
| $P^{t,i}_{ESS,ch}/P^{t,i}_{ESS,dch}$ | Charge/Discharge [MW] of $i^{th}$ participant with ESS at time t |
| $Q^{t,i}_{ESS}$ | Reactive power [Mvar] of $i^{th}$ participant with ESS at time t |
| $P^{t,i}_{ch}/P^{t,i}_{dch}$ | Charge/Discharge [MW] of $i^{th}$ at time t |
| $P_g/Q_g$ | Active / Reactive power [MW/Mvar] Generation |
| $\theta/v$ | Nodal voltage angle/magnitude [pu./rad] |
| $\Delta\theta/\Delta v$ | Nodal voltage angle/magnitude deviation [pu./rad] |
| $P^{t}_{WEM,buy}$ | Power purchases [MW] from WEM at time t |
| $P^{t}_{VPP,buy}/P^{t}_{VPP,sell}$ | VPP's Power purchases/sales from WEM at time t |
| $P_{DER}/Q_{DER}$ | Active/Reactive power output [MW/Mvar] of DER |
| $SOC^{t,i}_{ESS}$ | SOC [%] of $i^{th}$ ESS node at time t |
| $SOC^{t,i}_{MG}$ | SOC [%] of $i^{th}$ MG node at time t |
| $SOC^{ini,i}_{SS}/SOC^{end,i}_{SS}$ | Initial/end time SOC [%] of $i^{th}$ participant for SS |
| $SOC^{ini,i}_{ESS}/SOC^{end,i}_{ESS}$ | Initial/end time SOC [%] of $i^{th}$ participant with ESS |
| $P^{t,i}_{LEM,match}$ | Matching amount [MW] on LEM $i^{th}$ participant at time t |
| $SOC^{t,i}_{SS}$ | SOC [%] of $i^{th}$ participant with ESS at time t for ss |
| $D^{t}_{LEM,buy}/S^{t}_{LEM,sell}$ | Sum of buying/selling amount of buyer/seller |
| $C^{t,i,*}_{exp}$ | Self-scheduling expected cost with uncertainty |
| $C^{t,*}_{WEM}$ | WEM purchase cost of DSO with uncertainty |
| $C^{t,*}_{VPP}$ | WEM bidding cost of VPP with uncertainty |
| $C^{t,*}_{LFM}$ | LFM cost of DSO with uncertainty |

*Binary variables*

| | |
|---|---|
| sub | Binary integer variable of node relation |
| $b^{\theta}/b^{v}$ | Binary integer vector associated $\Delta\theta/\Delta v$ |
| $\alpha^{t,i}_{seller,PV}$ | Binary variable of $i^{th}$ PV state at time t |
| $\alpha^{t,i}_{seller,MG}$ | Binary variable of $i^{th}$ MG state at time t |
| $\alpha^{t,i}_{seller,ESS}$ | Binary variable of $i^{th}$ ESS state at time t |
| $\mu^{t,i}_{SC,ch}/\mu^{t,i}_{SC,dch}$ | $i^{th}$ node charging/discharging state at time t for SS |
| $\mu^{t,i}_{ESS,ch}/\mu^{t,i}_{ESS,dch}$ | $i^{th}$ node ESS charging/discharging state at time t for SS |

*Random variables*

| | |
|---|---|
| $P^{t,i}_{d}/Q^{t,i}_{d}$ | Power demand [MW/Mvar] of $i^{th}$ node at time t $\in \mathbf{P_d}/\mathbf{Q_d}$ |
| $P^{t,i}_{PV}/Q^{t,i}_{PV}$ | Power demand [MW/Mvar] of $i^{th}$ PV node at time t |
| $\varepsilon_{PV}$ | Uncertainty of PV |
| $\varepsilon_{P_d}, \varepsilon_{Q_d}$ | Uncertainty of active/reactive power demand |

*Indices and parameters*

| | |
|---|---|
| t | Index of Scheduling time steps [1hour] |
| T | Total number of scheduling time in one day |
| i, j | Node number index |
| k, l | Line number index |
| m | Piecewise linearization parameter $\in [1, M]$ |
| $b_c$ | Charge susceptance |
| $\tau$ | Per unit tap ratio |
| $\theta_s$ | Transformer shift angle |
| $\epsilon_{LEM}$ | Convergence coefficient for LEM clearing |
| $n_b/n_l$ | Total Number of buses/lines |
| $n_{VPP}$ | the Number of VPP Participants |
| $n_{DER}$ | the Number of DER |
| $n_{GL}$ | the Number of GL |
| $n_{MG}$ | the Number of MG |
| $n_{ESS}$ | the number of ESS Participants |
| BM | Big-M constant |
| $\eta_{ch}/\eta_{dch}$ | Coefficient of charging/discharging efficiency |
| $C^{i}_{ESS}$ | Rated capacity [MWh] of battery of $i^{th}$ node |
| $Y_b$ | Nodal admittance matrix $\in \mathbb{Z}^{NB\times NB}$ |
| $Y'_b$ | Adjusted nodal admittance matrix $\in \mathbb{Z}^{NB\times NB}$ |
| $Y_f/Y_t$ | Branch from/to matrices $\in \mathbb{Z}^{NL\times NB}$ |
| $Y'_f/Y'_t$ | Adjusted branch from/to matrices $\in \mathbb{Z}^{NL\times NB}$ |
| y | Complex series admittance |
| $y_{sh}$ | shunt admittance |
| $y^{i}_{ff}, y^{i}_{ft}, y^{i}_{tf}, y^{i}_{tt}$ | Admittance parameters |
| $y'^{i}_{ff}, y'^{i}_{tt}$ | Adjusted admittance parameter |
| g/b | Per unit line conductance / susceptance vector $\in \mathbb{Z}^{NL\times 1}$ |
| $\pi^{t}_{LEM,ini}$ | Initial clearing price [cent/kWh] of LEM |
| $P^{t,L_k}_{LSF}/Q^{t,L_k}_{LSF}$ | LSF of $k^{th}$ line at time t |
| $P^{t,L_k}_{Loss}$ | Line active power loss [MW] of $k^{th}$ line at time t |
| $\pi^{t,i}_{LFM,up}/\pi^{t,i}_{LFM,dn}$ | Up/Down flexibility price [cent/kWh] of $i^{th}$ participant at time t |
| $\pi^{t}_{exp}$ | Expected price [cent/kWh] of LEM at time t |
| $\pi^{t}_{WEM,buy}/\pi^{t}_{WEM,sell}$ | WEM bidding price [cent/kWh] of $i^{th}$ node at time t |
| $C_f/C_t/C_{br}$ | Node branch incidence matrix $\in \mathbb{Z}^{NL\times NB}$ |
| $C_g$ | Generator connection matrix |
| $k_1/k_2$ | Line loss coefficient from phase/voltage variation |
| $P_f/Q_f$ | Active/Reactive power flow through branch |
| $A_q$ | Piecewise constant for line capacity constraint |
| $S^{L}_{rate}$ | Line capacity limit [MVA] |
| $S^{t,i}_{rate}$ | Rated output power [MVA] of $i^{th}$ node at time t |
| $\sigma_n$ | Variance of distribution |
| $\boldsymbol{X}, \boldsymbol{X}^{*}$ | Historical/target input for gaussian process |
| $\boldsymbol{Y}$ | Joint distribution |
| $\boldsymbol{K}$ | Kernal function |
| $C^{t}_{WEM}$ | Cost for buying power from WEM at time t |
| $C^{t}_{VPP}$ | VPP operating cost at time t |
| $C^{t}_{LFM}$ | LFM cost at time t |
| $P^{t,i}_{flex,up}/P^{t,i}_{flex,dn}$ | Up/Down flexibility from $i^{th}$ participant at time t |
| $P^{t,i}_{LEM,buy,mis}$ | LEM power buy mismatch of $i^{th}$ participant at time t |
| $P^{t,i}_{LEM,sell,mis}$ | LEM power sell mismatch of $i^{th}$ participant at time t |
| $a_m, b_m, c_m$ | Output power piecewise linearization constant |

| | | | |
|---|---|---|---|
| $P^{t,i}_{PV,ub}$ | Upper bound of PV generation [MW] of $i^{th}$ node | $P^{t,i}_{d,ub}$ | Upper bound of power demand [MW] of $i^{th}$ node at time t |
| $P^{t,i}_{PV,lb}$ | Lower bound of PV generation [MW] of $i^{th}$ node | $P^{t,i}_{d,lb}$ | Lower bound of power demand [MW] of $i^{th}$ node at time t |
| $V_{max}/V_{min}$ | Maximum/minimum voltage | $\boldsymbol{X}_{PV}$ | Input dataset of PV for uncertainty forecasting |
| $V_{buff}$ | Voltage buffer coefficient | $\boldsymbol{X}_{d}$ | Input dataset of demand for uncertainty forecasting |

participant uncertainty. It has also been proposed to improve the revenue of the VPP while responding to the generation uncertainty of resources participating in the VPP through day-ahead scheduling and real-time operation [8–10].

However, there are several challenges that are currently encountering to activate VPPs. First, VPPs need to perform fair profit allocation with participants, but this requires complex methods such as game theory and dynamic pricing [11]. In addition, VPPs need to consider solving problems in the connected distribution network in cooperation with the DSO during the bidding process. When a VPP bids into the energy market with other generators, it sometimes does not consider the conditions of the distribution network to which the resources are connected. Moreover, the VPP is subject to a balancing penalty when resource uncertainty occurs after bidding in the day-ahead market [12]. Therefore, in this paper, a coordination framework of VPPs and DSOs that solves these problems is discussed. The analysis of related works is detailed in the next section.

### *1.1. Related works.*

Before discussing VPPs, the research cases on the management of DERs by DSOs are reviewed. From an impartiality perspective, a suitable approach to manage transmission constraints and voltage instability at the distribution network level is the application of local flexibility markets (LFM) [13]. In [14], a framework for LFM is proposed based on the mixed integer linear programming (MILP) method through a platform that aggregates flexibility resources called smart energy service provider. Market structures for aggregators and DSOs to coordinate to aggregate flexibility resources are presented in [15,16]. In addition, in [17], a model in which DSOs operate LFMs to aggregate flexibility resources in multiple timelines is proposed. There are also cases where flexibility through ESS is considered in association with business models such as peer-to-peer energy trading [18]. On the other hand, there are also cases that consider the uncertainty of participants in the operation of LFMs. In [19], a model is proposed in which an aggregator participates in a LFM in an energy community considers uncertainties on three factors: price, PV production, and load. Also, in [20], a LFM operated by the DSO is proposed to consider the probability of congestion caused by the load to provide flexibility through LFM. Moreover, it has been proposed to operate LFM through a two-stage stochastic model based on AC optimal power flow (ACOPF) and demand uncertainty [21].

Sometimes VPPs play the same role that the LFM perform. VPPs contribute to the stability of the power system by considering network constraints in their bidding process or by operating in coordination with DSOs. The most basic way to achieve optimal operation of VPP, including conditions in the distribution system, is to organize the technical constraints of the VPP based on a DC optimal power flow (DCOPF) model. In [22], a framework is proposed in which VPPs bid to account for the uncertainty of grid constraints and DSOs are incentivized to satisfy the constraints. There are cases where VPPs provide ancillary services to the grid by supplying active and reactive power through DERs [23,24]. In addition, [25] validated a bi-level cooperative operation scheme in which DSOs adjust their costs when trading power with VPPs at the upper level to ensure flexibility, and VPPs adjust their resources at the lower level. There are also cases where the VPP manages the LFM directly. In [26], a paper was presented to derive the optimal operating and risk-averse costs for the uncertainty of RES generation and load by utilizing the P robust model and fuzzy satisfaction approach based on DC power flow. Similarly, the operation of VPP based on interval optimization by applying DC power flow was discussed [27]. On the other hand, in distribution systems with short lines, the DC power flow model is unsuitable because of the large resistance of the lines, and it is difficult to consider voltage stability, which is essential in distribution systems because it does not consider the flow of reactive power [28]. In [29], the case of a VPP operated flexibility market directly, considering active and reactive power as linearized constraints, is shown, but losses are not considered. In [30], the case of a VPP operated flexibility market directly, considering active and reactive power as linearized constraints, is shown, but losses are not considered. A different approach [31] considers active and reactive power based on capacity, but similarly, losses are not considered. In [32], an iterative approach is used to consider non-linear network constraints, but this has the potential to be non-optimal. There is also research that considers losses due to VPP behavior as a quadratic-based approximation model [33].

In addition to coordination with the DSO, the VPP's interaction with the participants is also an important consideration. The VPP generally formulates problems with the objective of achieving aggregate profits without considering the profits of the participants. In this case, the VPP gets the best profit, but the participants may not. To solve this problem, research is being conducted to consider both the profit of the VPP platform and the profit of the participants through the game theory method or multi-stage method. The use of game theory to consider both the profit of VPP objects and the profit of participants has been discussed in [34,35]. In [36] and [37], a bi-level method is proposed to construct a upper-market bidding strategy for VPP by recruiting the results of cost minimizing scheduling for each VPP participant. In the situation of multiple competing VPPs, methods to improve the profits of each participant are discussed in [38]. In [39], a coordination between VPPs and electric vehicle charging stations is proposed to maximize the overall profit based on the incentives for each participant. There are also examples that combine both approaches. In [40], a two-level framework is proposed for VPPs participating in diverse markets, where each participant's profit is distributed based on a Stackelberg game.

Furthermore, VPPs are required to consider uncertainty in the day-ahead market bids to avoid imbalance penalty. For example, [41] proposes a method for ED settlement inside a VPP that can respond to generation uncertainty of renewable energy while maintaining the privacy of multiple DER participants in the VPP, and [42] proposes a method for renewable energy generators participating in a VPP to utilize a P2P market inside the VPP to respond to generation uncertainty. In [43], it is discussed how VPPs participating in day-ahead energy market, carbon emission market, and gas market operate considering the uncertainty of renewable energy. There is also a case study in [44] that proposed a VPP operation strategy that considers uncertainty based on scenarios and analyzes the benefits of each participant [45]. In addition, [45] applies a two-stage robust optimization approach by estimating correlations through data-driven uncertainty modeling. ADMM-based distributed optimization methods have also been proposed to consider the day-ahead bidding problem with resource uncertainty and network constraints [46].

Table 1 summarizes the previous works by several features. Many of the literatures on VPPs consider resource uncertainty and network constraints. However, in practice, VPPs do not have access to grid information and need to consider it through coordination with DSOs. Furthermore, most of the literature interacts with the participants to

**Table 1**
Reference classification for virtual power plant.

| Reference No. | Features/conditions considered | | | | |
|---|---|---|---|---|---|
| | Market Term | Voltage Stability | Uncertainty | Market option | VPP-DSO Coordination |
| [5,7,41–43,45] | Day-Ahead | X | O | X | X |
| [8,9] | Day-Ahead/Real-time | | | | |
| [22,35,40,44] | Day-Ahead | X | O | X | X |
| [10,34,36–39] | Day-Ahead/Real-time | | | | |
| [23,31] | Day-Ahead | O | X | X | X |
| [24,32] | Day-Ahead | O | X | X | O |
| [26] | Day-Ahead | O | O | X | X |
| [27 30] | Day-Ahead/Real-time | | | | |
| [22,25,46] | Day-Ahead | O | O | X | O |
| [29,33] | Real-Time | O | O | X | O |
| **Proposed Method** | **Day-Ahead** | O | O | O | O |

distribute profits, but in some cases, this may lead to the penalty of the VPP operator. Therefore, in this paper, a framework that overcomes this problem is discussed.

### 1.2. Contributions of research

In this paper, hierarchical robust day-ahead VPP and DSO coordination framework based on local market is proposed as a way to solve the problems discussed in the previous section. Since the markets considered in this paper are various, it is difficult to consider them in a centralized problem. Also, there is a privacy concern if each resource is not optimized individually. Therefore, the framework applied in this paper is a hierarchical optimization. Before the VPP bids, DERs are free to trade in the DSO operated LEM. DERs use gaussian process (GP) to forecast renewables and load with uncertainty and perform self-scheduling based on expected prices under worst-case conditions. The scheduling result determines the state of the seller or buyer, and free trading is performed within the sell and buy prices of the wholesale energy market (WEM). The LEM is cleared through a double auction mechanism. Mismatches that occur in the LEM are aggregated by VPPs to participate in the day-ahead WEM managed by the TSO. For this purpose, LFM operated by the DSO is applied. VPPs are granted a minimum of profits from their adjusted bids, and VPP are compensated for supplying flexibility through the adjustment with a loss-sensitivity-based flexibility price. In the DSO's operation scheduling, the network constraints are considered in a robust model based on MIP-OPF with linearized network constraints and worst-case scenarios of load and DER's generation. The proposed framework is validated through Monte-Carlo simulation (MCS) based scenario generation on IEEE benchmark system in MATLAB environment. Compared to existing research, the contributions of the framework proposed in this paper are as follows:

1) DERs are provided with a variety of market options. After the DER performs direct trades between participants through LEMs within the distribution network, the VPP aggregates the mismatched amounts in the LEMs and obtains revenues through WEM and LFM.
2) To consider the profits of both the VPP and the participants, it applies a hierarchical structure of individual and centralized problems. This structure can also contribute to the privacy of participants and the growth of DER.
3) In the LFM, the VPP are compensated with a loss sensitivity-based flexibility price when rescheduling is performed by the DSO due to network stability. Therefore, it may be applied as an appropriate compensation when the behavior is different from the previous scheduling.
4) A robust optimization in the form of MILP with the worst-case scenario considered step by stage is applied. In LEM, maximum net load is worst. In the case of VPP-DSO coordination, the maximum and minimum net load is considered as worst for voltage stability.

The remainder of this paper is organized as follows. Section 2 discusses the framework of the proposed VPP-DSO coordination. Section 3 defines uncertainty sets modeling and the robust optimization problem of each stage. Section 4 verifies the configured simulation environment and proposed method in the IEEE benchmark system. Finally, Section 5 presents the conclusion of the study, and discusses the advantages of the proposed method, limitations, and future works.

## 2. Proposed VPP-DSO coordination framework

The VPP model considered in the paper is a tariff-base model as a player operating for commercial purposes. In a general VPP, the participant's resources are managed by the VPP to participate in the energy market. However, in the LEM model, DERs prefer not to participate in VPPs. This is because participation in the LEM, with its relatively high expected profit, is more beneficial than participation in the upper market through VPPs. Therefore, in this paper, the DER's profit is ensured through the LEM and the VPP is aggregating mismatching amounts, and the VPP can get additional profit for coordination through the LFM and WEM. This approach enables VPP participants and VPPs to increase profits and DSOs to ensure stable system operations. It is assumed that generation and load forecasting, scheduling, and coordination for these processes are considered services provided to participants from VPP. DERs on the distribution grid are free to trade in the local energy market before the VPP performs aggregation. Each resource sometimes acts as a seller and sometimes as a buyer, depending on load conditions. The VPP aggregates the resulting mismatch and bids into the day-ahead energy market operated by the TSO. In general, VPPs participate in the reserve market operated by the TSO, but in this paper, the focus is on voltage enhancement in the distribution network, so only bidding in the day-ahead WEM is considered. In addition, the DSO's operation adjusts the scheduling of VPP participants to provide flexibility to improve voltage stability, and the VPP is compensated for providing flexibility with a flexibility price. The proposed framework and the interaction of participants in each market are depicted in Fig. 1 and Fig. 2.

### 2.1. DER's self-scheduling

In this paper, there are four types of participants in the distribution network: PV, ESS and Microgrid (MG). All participants perform the roles of seller and buyer. Furthermore, the LEM approach applied in this paper is like P2P energy trading. In the case of P2P energy trading, the initial strategy is based on the expected price from the WEM price, which is determined in advance. Therefore, this paper also assumes the same situation [47–49]. First, PV can only perform the role of seller if there is a surplus when PV is generating. This is defined by Eq (1).

$$\alpha_{seller,PV}^{t,i} = \begin{cases} 1, P_{PV}^{t,i} \geq P_d^{t,i} \\ 0, P_{PV}^{t,i} < P_d^{t,i} \end{cases} \tag{1}$$

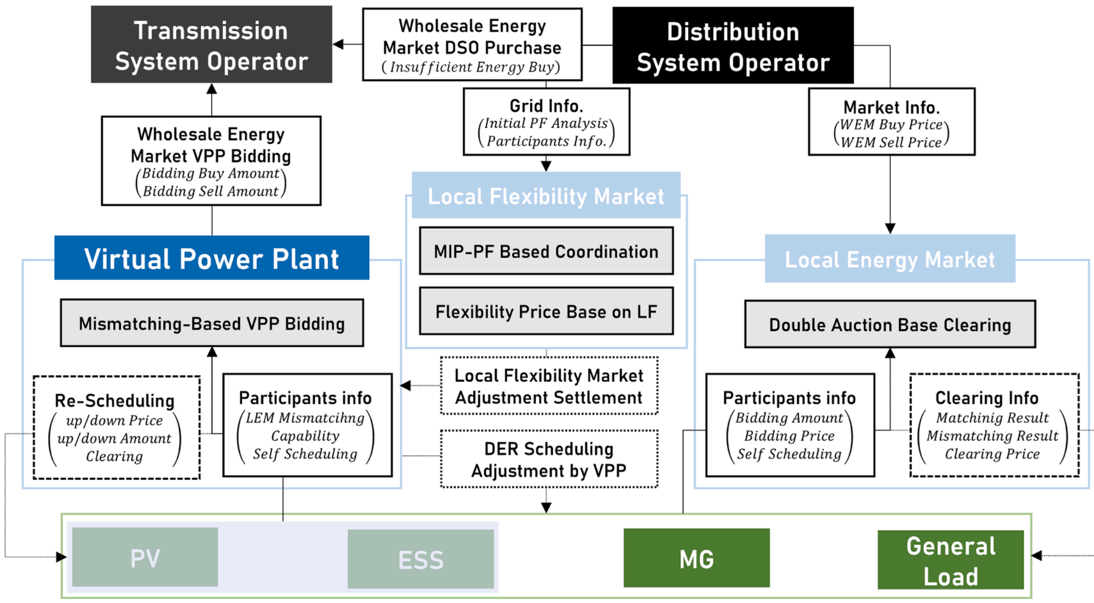


**Fig. 1.** Framework of proposed VPP-DSO coordination.

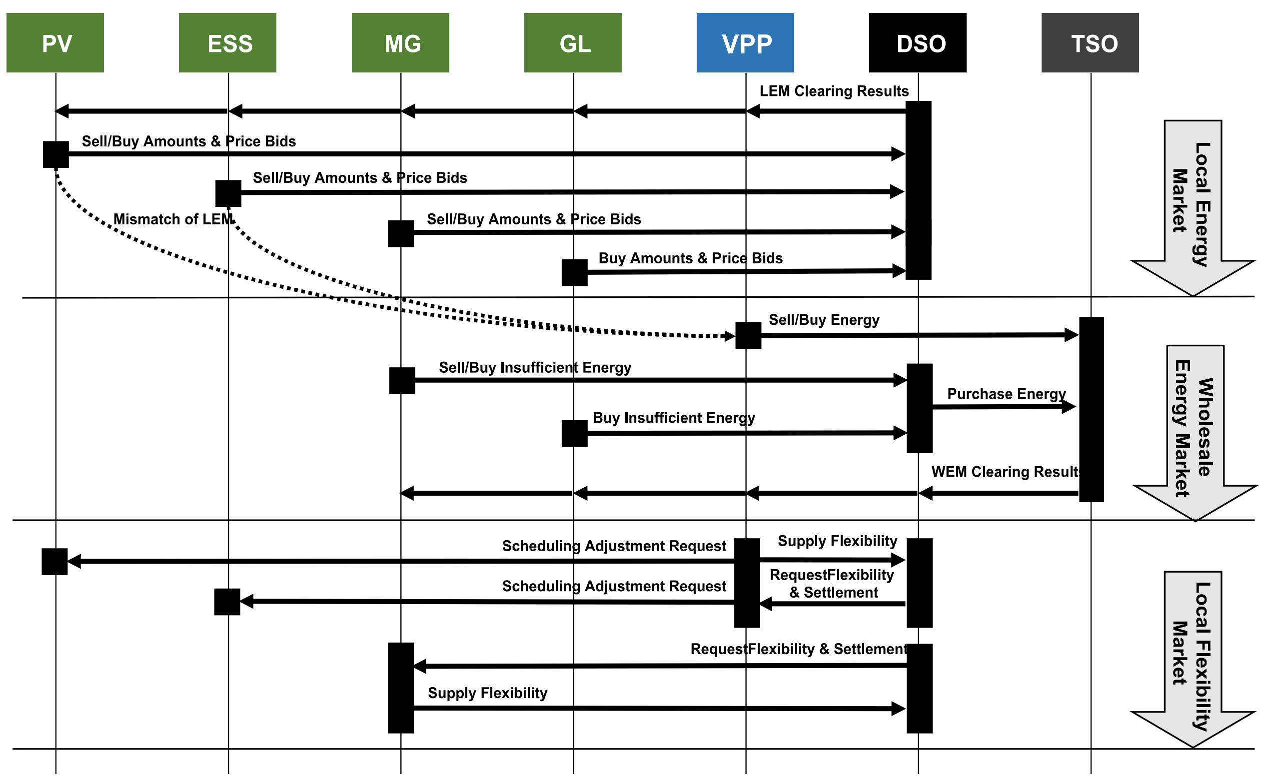


**Fig. 2.** Interaction of participants by market in the proposed framework.

On the other hand, for participants with ESS, it is difficult to determine the seller state because the price is not determined. Therefore, the seller state is determined after performing self-scheduling by applying the middle value of the sale and purchase price in the WEM as the expected price. The definition of the expected price is expressed in Eq. (2).

$$\pi_{exp}^{t} = \frac{\left(\pi_{WEM,buy}^{t} + \pi_{WEM,sell}^{t}\right)}{2} \tag{2}$$

The objective function and constraints for resource with ESS are implemented as a mixed integer linear programming (MILP) problem as follows:

$$min\sum\nolimits_{t}^{T}\left(P_{exp,buy}^{t,i} - P_{exp,sell}^{t,i}\right)\pi_{exp}^{t} \tag{3}$$

$S.t$

$$P_{exp,buy}^{t,i} - P_{exp,sell}^{t,i} + P_{SC,dch}^{t,i} - P_{SC,ch}^{t,i} = P_{d}^{t,i} - P_{PV}^{t,i} \tag{4}$$

$$SOC_{SC}^{t,i} - \frac{\eta_{ch}}{C_{ESS}}P_{SC,ch}^{t,i} + \frac{1}{\eta_{dch}{}^{*}C_{ESS}}P_{SC,dch}^{t,i} = SOC_{SC}^{t-1,i} \tag{5}$$

$$SOC_{SC}^{end,i} = SOC_{SC}^{ini,i} \tag{6}$$

$$\mu_{SC,ch}^{t,i} + \mu_{SC,dch}^{t,i} \leq 1 \tag{7}$$

$$\begin{cases} P_{SC,ch}^{t,i} - \mu_{SC,ch}^{t} S_{rate}^{t,i} \leq 0 \\ P_{SC,dch}^{t,i} - \mu_{SC,dch}^{t} S_{rate}^{t,i} \leq 0 \end{cases} \tag{8}$$

After self-scheduling, the seller state of the MG and ESS is determined as follows:

$$\alpha_{seller,MG}^{t,i} = \begin{cases} 1, P_{exp,buy}^{t,i} - P_{exp,sell}^{t} < 0 \\ 0, P_{exp,buy}^{t,i} - P_{exp,sell}^{t} \geq 0 \end{cases} \tag{9}$$

$$\alpha_{seller,ESS}^{i,t} = \begin{cases} 1, -P_{ch}^{t} + P_{dch}^{t} \geq P_{d}^{t} \\ 0, -P_{ch}^{t} + P_{dch}^{t} < P_{d}^{t} \end{cases} \tag{10}$$

### 2.2. Local energy market clearing

In LEM, matching is performed with all sellers depending on the seller state and sales amount determined by self-scheduling. The sale price and buy price are freely bid within the energy sale price and buy price of the WEM. The clearing of the LEM is achieved through a double auction mechanism. The flowchart of LEM clearing is shown in Fig. 3.

First, the initial clearing price is set as the average value of the buyer's lowest bid and the seller's highest bid.

$$\pi_{LEM,ini}^{t} = \frac{\left(\min(\pi_{LEM,buy}^{t,i}\right) + \max\left(\pi_{LEM,sell)}^{t,i}\right)\Big)}{2} \tag{11}$$

Next, if $D_{LEM,buy}^{t}$, the sum of the amounts of buyers buying above the initial clearing price is greater than $S_{LEM,sell}^{t}$, the sum of the amounts of sellers selling above the initial clearing price, the clearing price is increased, and vice versa, the price is decreased. If the two values are within the convergence condition, a price adjustment is done. The formula for the convergence condition is as follows:

$$\left|\left(D_{LEM,buy}^{t} - S_{LEM,sell}^{t}\right)\right| \leq \epsilon_{LEM} \tag{12}$$

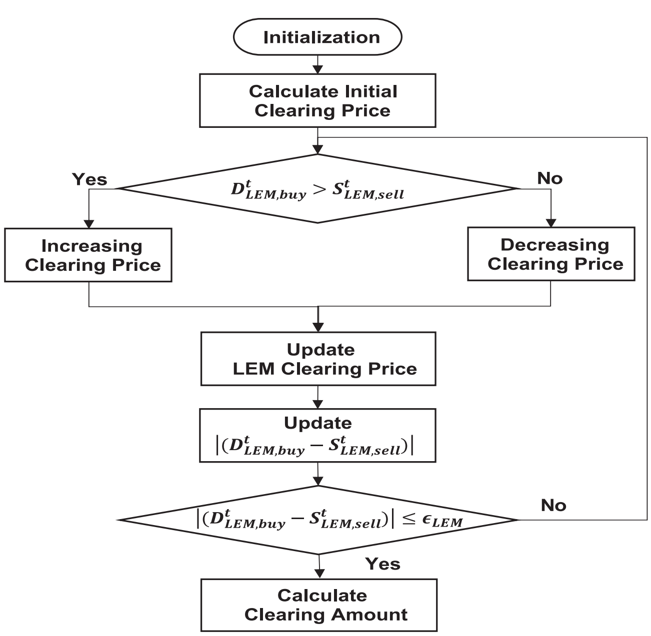


**Fig. 3.** Double auction mechanism-based lem clearing.

### 2.3. VPP and DSO coordination in LFM

The VPP-DSO coordination model proposed in the paper is configured as a centralized problem. The VPP participates in the WEM by aggregating mismatches from the LEM, with PV and ESS. Since MGs are supposed to self-consume as much as possible, they are considered competitors with the VPP in the DSO's coordination. In addition, MG's surplus power is traded with the DSO. The DSO then operates an LFM with a loss-sensitivity-based flexibility price to coordinate the scheduling of VPPs and MGs. In other words, the VPP bids on the WEM adjusted by the DSO to obtain profits, and the VPP are compensated by the DSO for supplying flexibility. Eqs. (13) − (17) show the process of determining the loss sensitivity. More details on loss sensitivity are described in [50]. Loss sensitivity is based on the branch power flow. The branch power flow is illustrated in Fig. 4.

$$sub(i,j) = \begin{cases} 1, \ \textit{if node j belongs to sub} - \textit{tree of node i} \\ \qquad 0, \ \textit{otherwise} \end{cases} \tag{13}$$

$$\frac{\partial P_{k}^{t}}{\partial P_{D}^{t,i}} = sub(k+1,i) + \sum_{j=k+2}^{NB} sub(k+1,j)*\frac{\partial P_{Loss}^{t,(j-1)}}{\partial P_{D}^{t,i}} \tag{14}$$

$$\frac{\partial Q_{k}^{t}}{\partial P_{D}^{t,i}} = \sum_{j=k+2}^{NB} sub(k+1,j)*\frac{\partial Q_{Loss}^{t,(j-1)}}{\partial P_{D}^{t,i}} \tag{15}$$

$$\frac{\partial P_{k,Loss}^{t}}{\partial P_{D}^{t,i}} = \left(2P_{k}^{t}\frac{\partial P_{k}^{t}}{\partial P_{D}^{t,i}} + 2Q_{k}^{t}\frac{\partial Q_{k}^{t}}{\partial P_{D}^{t,i}}\right)*\frac{r_k}{(V_{k+1}^{t})^2} \tag{16}$$

$$\frac{\partial P_{Loss}^{t}}{\partial P_{D}^{t,i}} = \sum_{k=1}^{NL} \frac{\partial P_{Loss}^{t,k}}{\partial P_{D}^{t,i}} \tag{17}$$

As a result, the sensitivity-based flexibility price is expressed as follows:

$$\pi_{LFM,dn}^{t,i} = \pi_{WEM,buy}^{t}\left(1 + \frac{\partial P_{Loss}^{t}}{\partial P_{D}^{t,i}}\right) \tag{18}$$

$$\pi_{LMF,up}^{t,i} = \pi_{WEM,buy}^{t}\left(1 - \frac{\partial P_{Loss}^{t}}{\partial P_{D}^{t,i}}\right) \tag{19}$$

Eq. (18) refers to the price for increasing power consumption by the bus/time step, and Eq. (19) is the price for decreasing power consumption. As explained above, $\frac{\partial P_{Loss}^{t}}{\partial P_{i,D}^{t}}$ refer to the variation of losses due to an increase in load, so the sign of $\frac{\partial P_{Loss}^{t}}{\partial P_{i,D}^{t}}$ in Eqs. (18)-(19) is opposite to each other.

In this paper, linearized network constraints are considered to account for the effects of the distribution network. However, linearized network constraints have the disadvantage of not being able to accurately account for line losses. To compensate for this limitation, MIP-PF is applied. MIP-PF improves the accuracy of the voltage because it considers line losses that are approximated, but not exact, by the big-M method [51]. In MIP-PF, the nodal admittance matrix $Y_b$ and its adjusted version $Y_b'$ are used to represent the effect of power flow on voltage angle

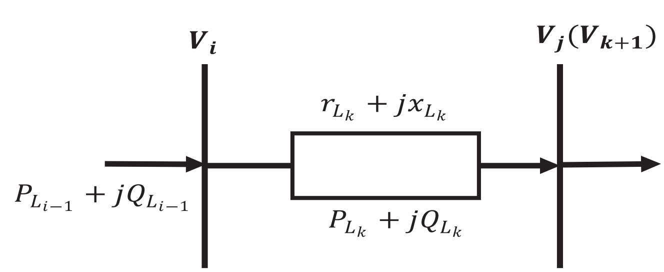


**Fig. 4.** Branch power flow of line $L_k$.

more accurately. (22)-(27) generalize the methodology to be applicable to any grid topology using the same notation as in [52].

$$Y_b = \begin{bmatrix} y_{ff}^i & y_{ft}^i \\ y_{tf}^i & y_{tt}^i \end{bmatrix} = \begin{bmatrix} (y + j\frac{b_c}{2})\frac{1}{\tau^2} & -y\frac{1}{\tau e^{-\theta_s}} \\ -y\frac{1}{\tau e^{\theta_s}} & (y + j\frac{b_c}{2}) \end{bmatrix} \tag{20}$$

$$Y_b' = \begin{bmatrix} y_{ff}'^i & y_{ft}^i \\ y_{tf}^i & y_{tt}'^i \end{bmatrix} = \begin{bmatrix} y\frac{1}{\tau e^{-\theta_s}} & -y\frac{1}{\tau e^{-\theta_s}} \\ -y\frac{1}{\tau e^{\theta_s}} & y\frac{1}{\tau e^{\theta_s}} \end{bmatrix} \tag{21}$$

$$Y_f = diag\left\{y_{ff}^1, \cdots, y_{ff}^{n_l}\right\}C_f + diag\left\{y_{ft}^1, \cdots, y_{ft}^{n_l}\right\}C_t \tag{22}$$

$$Y_t = diag\left\{y_{tf}^1, \cdots, y_{tf}^{n_l}\right\}C_f + diag\left\{y_{tt}^1, \cdots, y_{tt}^{n_l}\right\}C_t \tag{23}$$

$$Y_f' = diag\left\{y_{ff}'^1, \cdots, y_{ff}'^{n_l}\right\}C_f + diag\left\{y_{ft}^1, \cdots, y_{ft}^{n_l}\right\}C_t \tag{24}$$

$$Y_t' = diag\left\{y_{tf}^1, \cdots, y_{tf}^{n_l}\right\}C_f + diag\left\{y_{tt}'^1, \cdots, y_{tt}'^{n_l}\right\}C_t \tag{25}$$

$$Y_b = C_f^T Y_f + C_t^T Y_t + diag\{y_{sh}^1, \cdots, y_{sh}^{n_b}\} \tag{26}$$

$$Y_b' = C_f^T Y_f' + C_t^T Y_t' \tag{27}$$

$$\begin{bmatrix} -\Im\{Y_b'\} & \Re\{Y_b\} & -C_g & 0 \\ -\Re\{Y_b'\} & -\Im\{Y_b\} & 0 & -C_g \\ k_1 diag\{g\}|C_{br}|^T & k_2 diag\{g\}|C_{br}|^T & & \\ k_1 diag\{b\}|C_{br}|^T & k_2 diag\{b\}|C_{br}|^T & & \end{bmatrix} \begin{bmatrix} \theta \\ v \\ P_g \\ Q_g \\ \Delta\theta \\ \Delta v \end{bmatrix} = \begin{bmatrix} -P_d \\ -Q_d \end{bmatrix} \tag{28}$$

$$\begin{bmatrix} -\Im\left\{Y_f'\right\} & \Re\{Y_f\} \\ -\Re\left\{Y_f'\right\} & -\Im\{Y_f\} \end{bmatrix} \begin{bmatrix} \theta \\ v \end{bmatrix} = \begin{bmatrix} P_f \\ Q_f \end{bmatrix} \tag{29}$$

$$-BM\left(1 - b^\theta\right) \le -C_{br}\theta \le BMb^\theta \tag{30}$$

$$0 \le -C_{br}\theta + \Delta\theta \le 2BMb^\theta \tag{31}$$

$$0 \le C_{br}\theta + \Delta\theta \le 2BM(1 - b^\theta) \tag{32}$$

$$-BM(1 - b^v) \le -C_{br}v \le BMb^v \tag{33}$$

$$0 \le -C_{br}v + \Delta v \le 2BMb^v \tag{34}$$

$$0 \le C_{br}\theta + \Delta v \le 2BM(1 - b^v) \tag{35}$$

$$-S_{rate}^L \le P_f + A_q Q_f \le S_{rate}^L \tag{36}$$

$$-S_{rate}^L \le P_f - A_q Q_f \le S_{rate}^L \tag{37}$$

$$-S_{rate}^L \le A_q P_f + Q_f \le S_{rate}^L \tag{38}$$

$$-S_{rate}^L \le A_q P_f - Q_f \le S_{rate}^L \tag{39}$$

The MIP-PF is represented by the line loss coefficients $k_1$, $k_2$ from voltage and phase variation as shown in Eq. (28). The process of formulation is described in detail in [53]. Eq. (29) is the linearized branch flow equation, through which the amount of power flow on each line is calculated. To avoid the case that the optimal solution is not bounded by the line loss constraint, the big-M method is used, which is shown in Eqs. (30)-(35). In addition, the piecewise linearization method is applied to the line capacity constraint in Eqs. (36)-(39).

In conclusion, the DSO's operations are determined by the sum of the cost of buying power from the WEM and the cost of coordination. This includes the cost of the VPP's adjusted bid. Each cost is defined as follows:

$$C_{WEM}^t = P_{WEM,buy}^t \pi_{WEM,buy}^t \tag{40}$$

$$C_{VPP}^t = P_{Vpp,buy}^t \pi_{WEM,buy}^t - P_{Vpp,sell}^t \pi_{WEM,sell}^t \tag{41}$$

$$C_{LFM}^t = \sum_{i=1}^{n_{DER}} (\pi_{LFM,dn}^{t,i} P_{flex,dn}^{t,i} + \pi_{LFM,up}^{t,i} P_{flex,up}^{t,i}) \tag{42}$$

The sell and buy bids in the VPP are determined by the surplus of each resource and the coordinated scheduling. This is expressed as in Eqs. (36)-(37).

$$P_{Vpp,buy}^t - P_{Vpp,sell}^t = \sum\nolimits_{i=1}^{n_{VPP}} (P_{LEM,sell,mis}^{t,i} - P_{LEM,buy,mis}^{t,i} + -P_{flex,up}^{t,i} + P_{flex,dn}^{t,i}) \tag{43}$$

$$P_{WEM,buy}^t = \sum\nolimits_{i=1}^{n_{GL}} P_{LEM,buy,mis}^{t,i} + \sum\nolimits_{i=1}^{n_{MG}} (P_{LEM,buy,mis}^{t,i} - P_{flex,up}^{t,i} + P_{flex,dn}^{t,i}) \tag{44}$$

The flexibility that the VPP can supply depends on the resources that each participant has. First, PV only supplies downward flexibility. Furthermore, the reactive power supplied by PV is considered at a constant power factor $pf$. As a result, the output of PV is expressed as follows:

$$P_{PV}^{t,i} = P_{LEM,sell}^{t,i} + P_{VPP,sell}^{t,i} - P_{flex,dn}^{t,i} \tag{45}$$

$$Q_{PV}^{t,i} = P_{PV}^{t,i}\sqrt{1 - pf} \tag{46}$$

Participants with ESS are considered to make additional outputs to supply flexibility while maintaining their self-scheduling. Participants with ESS can adjust their output, providing both upward and downward flexibility. In coordination, the output of the ESS is determined as follows, which can be configured the same for MGs competing with VPPs.

$$P_{ESS,ch}^{t,i} - P_{ESS,dch}^{t,i} = P_{SC,ch}^{t,i} - P_{SC,dch}^{t,i} + P_{flex,dn}^{t,i} - P_{flex,up}^{t,i} \tag{47}$$

$$SOC_{ESS}^t - \frac{\eta_{ch}}{C_{ESS}} P_{ESS,ch}^{t,i} + \frac{1}{\eta_{dch} * C_{ESS}} P_{ESS,dch}^{t,i} = SOC_{ESS}^{t-1} \tag{48}$$

$$SOC_{ESS}^{end} = SOC_{ESS}^{ini} \tag{49}$$

$$\left(P_{ESS,ch}^{t,i} - P_{ESS,dch}^{t,i}\right)^2 + (Q_{ESS}^{t,i})^2 = S_{ESS,rate}^i \tag{50}$$

$$\mu_{ESS,ch}^{t,i} + \mu_{ESS,dch}^{t,i} \le 1 \tag{51}$$

$$\begin{cases} P_{ESS,ch}^{t,i} - \mu_{ESS,ch}^t S_{rate}^{t,i} \le 0 \\ P_{ESS,dch}^{t,i} - \mu_{ESS,dch}^t S_{rate}^{t,i} \le 0 \end{cases} \tag{52}$$

Considering the reactive power output of the ESS, as shown in Eq. (50), requires a very complex methodology because of its nonlinearity. Therefore, the piecewise linearization method is applied to approximate it. The piecewise linearization method is shown in Fig. 5. The capacity constraint of the ESS with the piecewise linearization method applied is expressed as follows:

$$a_m\left(P_{ESS,ch}^{t,i} - P_{ESS,dch}^{t,i}\right) + b_m Q_{ESS}^{t,i} \le c_m S_{ESS,rate}^i, m \in [1, L] \tag{53}$$

$$a_m = cos\frac{(-1 + 2m)\pi}{L}, b_m = sin\frac{(-1 + 2m)\pi}{L}, c_m = cos\frac{\pi}{L} \tag{54}$$

The linearized network constraint with the output of the DER is reconstructed as follows:

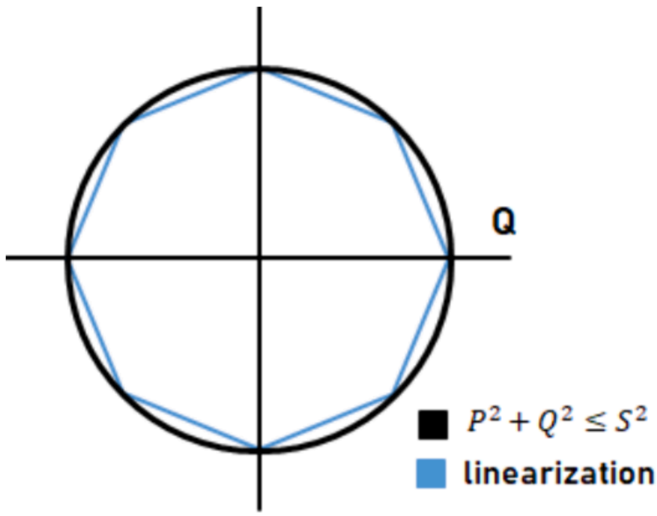


**Fig. 5.** Piecewise linearization method.

$$\begin{bmatrix} -\Im\{Y_b'\} & \Re\{Y_b\} & -C_g & 0 \\ -\Re\{Y_b'\} & -\Im\{Y_b\} & 0 & -C_g \\ k_1 diag\{g\}|C_{br}|^T & k_2 diag\{g\}|C_{br}|^T & & \\ k_1 diag\{b\}|C_{br}|^T & k_2 diag\{b\}|C_{br}|^T & & \end{bmatrix} \begin{bmatrix} \theta \\ v \\ P_g \\ Q_g \\ \Delta\theta \\ \Delta v \end{bmatrix} = \begin{bmatrix} -P_d \\ -Q_d \end{bmatrix} + \begin{bmatrix} P_{DER} \\ Q_{DER} \end{bmatrix} \tag{55}$$

Finally, the coordination problem of VPP and DSO is formulated as a MILP problem as follows:

$$Min \sum_{t=1}^{T} \left( C_{WEM}^t + C_{VPP}^t + C_{LFM}^t \right) \tag{56}$$

$$s.t.\ Eqs.\ (20) \ - \ (55)$$

## 3. Robust optimization formulation

### 3.1. Gp-based uncertainty forecasting

The uncertainties in DER and load considered in this paper are modeled based on gaussian distributions. In general, PV has output and uncertainty due to insolation and temperature. For loads, the uncertainty is due to the day of the week and its behavioral characteristics. The set of uncertainties modeled as a probability density function by a gaussian distribution is given below.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-u)^2}{2\sigma^2}} \tag{57}$$

In this paper, GP was applied as a method to generate the uncertainty set. GP-based forecasting is useful for time series data such as renewable energy generation and load, and is a useful method for forecasting nonlinear functions, helping to characterize the data and provide uncertainty bounds [54,55]. An example of the GP-based method is shown in Fig. 6. The results of the forecast using GP and the covariance using the K-kernel function are shown in Eqs. (58) and (59).

$$y = f(x) + \varepsilon, \varepsilon\ N(0, \sigma^2) \tag{58}$$

$$cov(f(x_i), f(x_j)) = k(x_i, x_j) \tag{59}$$

When predicting a function value $f_*(\boldsymbol{X}^*)$ for $\boldsymbol{X}^*$, the prior distribution is then organized as follows:

$$\begin{bmatrix} f(\boldsymbol{X}) \\ f_*(\boldsymbol{X}^*) \end{bmatrix} GP(0, \begin{bmatrix} K(\boldsymbol{X},\boldsymbol{X}) & K(\boldsymbol{X},\boldsymbol{X}^*) \\ K(\boldsymbol{X}^*,\boldsymbol{X}) & K(\boldsymbol{X}^*,\boldsymbol{X}^*) \end{bmatrix}) \tag{60}$$

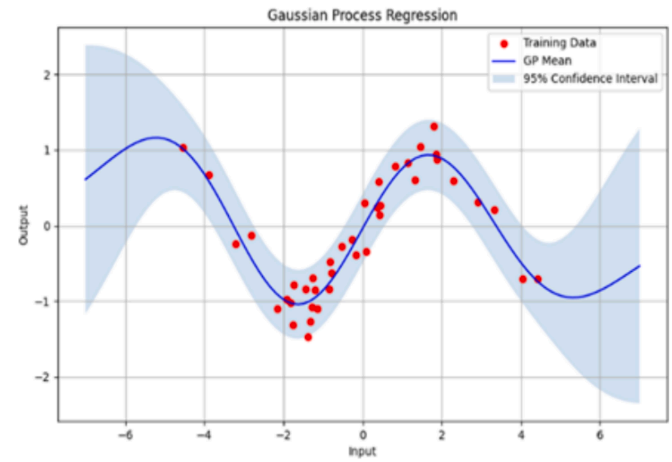


**Fig. 6.** Example of forecasting using GP.

$$K(\boldsymbol{X},\boldsymbol{X}) = \begin{pmatrix} K(x_1, x_1) & \cdots & K(x_1, x_n) \\ \vdots & \ddots & \vdots \\ K(x_n, x_1) & \cdots & K(x_n, x_n) \end{pmatrix} \tag{61}$$

With likelihood, the joint distribution of $\boldsymbol{Y}, f_*(\boldsymbol{X}^*)$, is obtained, and the posterior distribution of the function is follows:

$$\begin{bmatrix} \boldsymbol{Y} \\ f_*(\boldsymbol{X}^*) \end{bmatrix} GP(0, \begin{bmatrix} K(\boldsymbol{X},\boldsymbol{X}) + \sigma_n^2 I_n & K(\boldsymbol{X},\boldsymbol{X}^*) \\ K(\boldsymbol{X}^*,\boldsymbol{X}) & K(\boldsymbol{X}^*,\boldsymbol{X}^*) \end{bmatrix}) \tag{62}$$

Using the properties of the normal distribution, the conditional distribution is formed. This is expressed as Eq. (63).

$$f_*(\boldsymbol{X}^*)|\boldsymbol{Y}\ N(\overline{f_*}, \sigma^2(f_*)) \tag{63}$$

The mean and variance of a conditional distribution are defined as follows:

$$\overline{f_*} = K(\boldsymbol{X},\boldsymbol{X}^*)^T \left(K(\boldsymbol{X},\boldsymbol{X}) + \sigma_n^2 I_n\right)^{-1} \boldsymbol{Y} \tag{64}$$

$$\sigma^2(f)_* = K(\boldsymbol{X}^*,\boldsymbol{X}^*) - K(\boldsymbol{X}^*,\boldsymbol{X})\left(K(\boldsymbol{X},\boldsymbol{X}) + \sigma_n^2 I_n\right)^{-1} K(\boldsymbol{X},\boldsymbol{X}^*) \tag{65}$$

For PV, the input set consists of irradiance and temperature, and for load, the day of the week is used as an input to categorize commercial and residential. Based on the predicted mean and variance, cases within a 95 % confidence interval are applied. Therefore, the maximum and minimum of PV output and load are defined as follows:

$$\begin{cases} P_{Pv,lb} = f_{PV}(\boldsymbol{X}_{PV}) - \varepsilon_{PV} \\ P_{Pv,ub} = f_{PV}(\boldsymbol{X}_{PV}) + \varepsilon_{PV} \end{cases} \tag{66}$$

$$\begin{cases} P_{d,lb} = f_{P_d}(\boldsymbol{X}_d) - \varepsilon_{P_d} \\ P_{d,ub} = f_{P_d}(\boldsymbol{X}_d) + \varepsilon_{P_d} \end{cases} \tag{67}$$

$$\begin{cases} Q_{d,lb} = f_{Q_d}(\boldsymbol{X}_d) - \varepsilon_{Q_d} \\ Q_{d,ub} = f_{Q_d}(\boldsymbol{X}_d) + \varepsilon_{Q_d} \end{cases} \tag{68}$$

### 3.2. Reformulation of problem

The application of robust optimization is different for each stage. First, the process of the DER's self-scheduling is described. For the DER to bid into the local energy market, it considers the worst case with the lowest PV generation and the highest load. This is constructed by the GP-based forecast results. As a result, the optimization problem is reformulated as follows, keeping the MILP form.

$$min \sum_t^T \left( C_{exp}^{t,i,*} \right) \tag{69}$$

s.t

$$P^{t,i}_{exp,buy} - P^{t,i}_{exp,sell} + P^{t,i}_{SC,dch} - P^{t,i}_{SC,ch} = P^{t,i}_{d,lb} - P^{t,i}_{PV,ub} \quad (70)$$
$$Eqs.\ (5) - (8)$$

The Local Energy Market's clearing is based on the results of the robust optimization performed in the self-scheduling stage and when the load of the general participants is highest. The robust optimization is performed in the VPP-DSO coordination stage afterwards. If the voltage is deviated from the defined range, the power system becomes unstable. The minimum of the voltage is expected to arise when the net load is high and the maximum when the net load is low. Therefore, the network constraint, Equation (55), is reformulated when uncertainty is included as follows:

$$\begin{bmatrix} P^{t,i}_{d,lb} - P^{t,i}_{PV,ub} \\ Q^{t,i}_{d,lb} - Q^{t,i}_{PV,ub} \end{bmatrix} \le \begin{bmatrix} -\Im\{Y'_b\} & \Re\{Y_b\} & -C_g & 0 \\ -\Re\{Y'_b\} & -\Im\{Y_b\} & 0 & -C_g \\ k_1 diag\{g\}|C_{br}|^T & k_2 diag\{g\}|C_{br}|^T & & \\ k_1 diag\{b\}|C_{br}|^T & k_2 diag\{b\}|C_{br}|^T & & \end{bmatrix} \begin{bmatrix} \theta \\ v \\ P_g \\ Q_g \\ \Delta\theta \\ \Delta v \end{bmatrix}$$
$$= \begin{bmatrix} -P_d \\ -Q_d \end{bmatrix} + \begin{bmatrix} P_{DER} \\ Q_{DER} \end{bmatrix} \le \begin{bmatrix} P^{t,i}_{d,ub} - P^{t,i}_{PV,lb} \\ Q^{t,i}_{d,ub} - Q^{t,i}_{PV,lb} \end{bmatrix} \quad (71)$$

An additional consideration to increase the robustness of the coordination is to complement the linearized constraints. Although MIP-PF is a highly accurate solution, many linearization methods have some error. Therefore, a voltage buffer factor is also considered to construct a tighter voltage constraint. The equation for this is expressed as follows:

$$V_{min} + V_{buff} \le V^t_j \le V_{max} - V_{buff} \quad (72)$$

In addition, the DSO's WEM purchases and VPP's bids are also considered over a range in the uncertainty. The equation is expressed as follows:

$$P^t_{Vpp,buy} - P^t_{Vpp,sell} = \sum_{i=1}^{n_{VPP}} (P^{t,i}_{d,lb} - P^{t,i}_{PV,ub} - P^{t,i}_{LEM,match} - P^{t,i}_{flex,up} + P^{t,i}_{flex,dn}) \quad (73)$$

$$P^t_{WEM,buy} = \sum_{i=1}^{n_{MG}} (P^{t,i}_{d,lb} - P^{t,i}_{PV,ub} - P^{t,i}_{LEM,match} - P^{t,i}_{flex,up} + P^{t,i}_{flex,dn}) + \sum_{i=1}^{n_{GL}} (P^{t,i}_{d,lb} - P^{t,i}_{LEM,match}) \quad (74)$$

As a result, a robust optimization model for VPP-DSO coordination remains a MILP problem and is considered as an interval optimization. It is formulated as follows:

$$Min \sum_{t=1}^{T} (C^{t,*}_{WEM} + C^{t,*}_{VPP} + C^{t,*}_{LFM}) \quad (75)$$

$$s.t.$$
$$Eqs.\ (20) - (42)$$
$$Eqs.\ (45) - (54)$$
$$Eqs.\ (71) - (74)$$

The results of the optimization are validated by random scenarios generated based on the MCS. As a results, to summarize the flowchart of the overall process, it is depicted as Fig. 7.

## 4. Case study

### 4.1. Simulation setting

To validate the proposed framework, an IEEE 33 bus system [56] is configured as shown in Fig. 8. The capacities of the resources present in the distribution network are summarized in Table 2.

The data for GP-based forecasting are PV data for Berlin, Germany from PVGIS [57] and load data for California, USA from NREL [58]. The PV output and active and reactive power profiles for each bus with uncertainty ranges are shown in Figs. 9, 10, and 11.

Wholesale market prices were applied by randomly adjusting data from PJM within a certain range [59].

The validation of the proposed framework is performed for a total of four cases. Case 1 is the VPP controls all resources and only participates in the WEM. In this case, VPP participants are not allowed to participate in LEM, and the VPP bids without distribution network influence. In other words, coordination between the VPP and DSO is not considered. Case 2 shows that DERs are allowed to participate in LEM, but there is

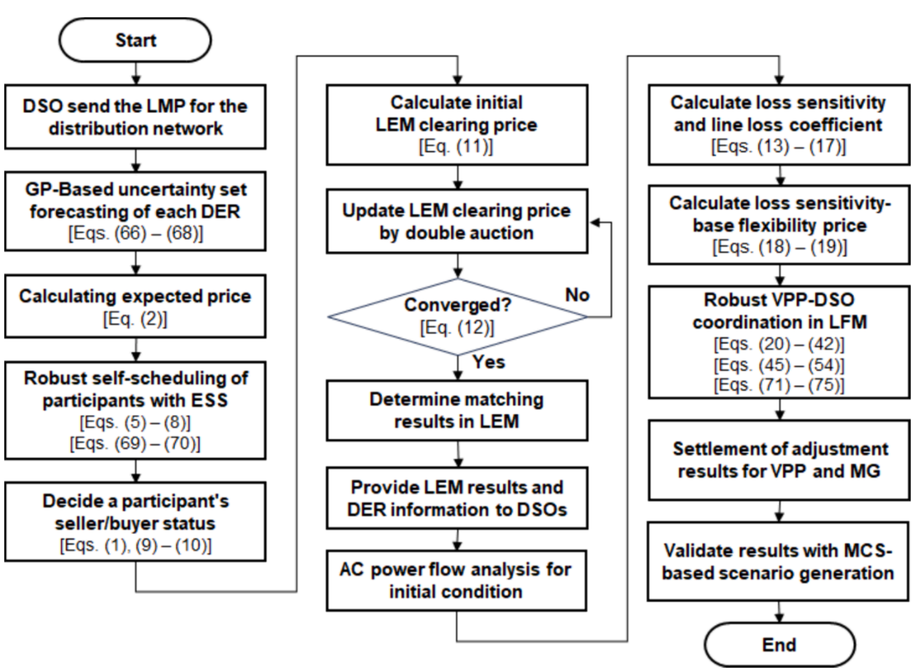


**Fig. 7.** Flowchart of overall process.

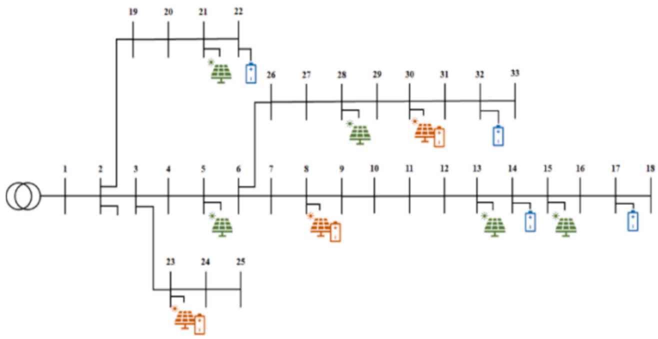


**Fig. 8.** Single line diagram of IEEE 33 bus system.

**Table 2**
Parameters of DERs.

| Bus No. | $C_{PV}$ [MW] | $C_{ESS}$ [MWh] | $S^{ESS}_{rate}$ [MW] | $SOC^{ESS}_{ini}$(%) | $\eta_{ESS}$(%) |
|---|---|---|---|---|---|
| 5 | 0.7 | – | – | – | – |
| 8 | 0.3 | 0.5 | 0.25 | 0.5 | 0.95 |
| 13 | 0.8 | – | – | – | – |
| 14 | – | 0.4 | 0.2 | 0.5 | 0.95 |
| 15 | 0.9 | – | – | – | – |
| 17 | – | 0.4 | 0.2 | 0.5 | 0.95 |
| 21 | 0.5 | – | – | – | – |
| 22 | – | 0.4 | 0.2 | 0.5 | 0.95 |
| 23 | 0.3 | 0.5 | 0.25 | 0.5 | 0.95 |
| 28 | 0.5 | – | – | – | – |
| 30 | 0.3 | 0.5 | 0.25 | 0.5 | 0.95 |
| 32 | – | 0.4 | 0.2 | 0.5 | 0.95 |

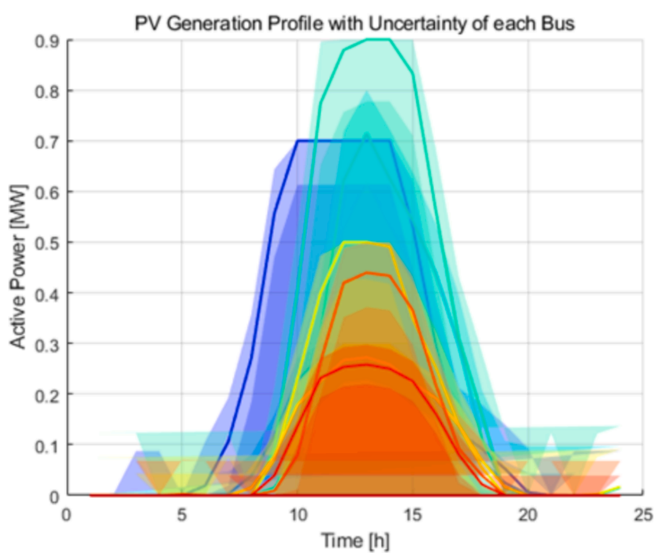


**Fig. 9.** PV profile with uncertainty of each bus.

still no coordination with the DSO. Case 3 represents the case where the LEM participation of VPP participants is restricted and only the coordination of DSO and VPP is considered. Finally, in Case 4, the framework proposed in the paper is applied. Each case is analyzed in detail from the perspective of voltage stability and participants' profits.

### 4.2. Self-scheduling of DERs

In Figs. 12 and 13, the results of self-scheduling for participants with ESS and MG participants are shown. Positive values mean discharge and negative values mean charge. Since the initial expected prices are

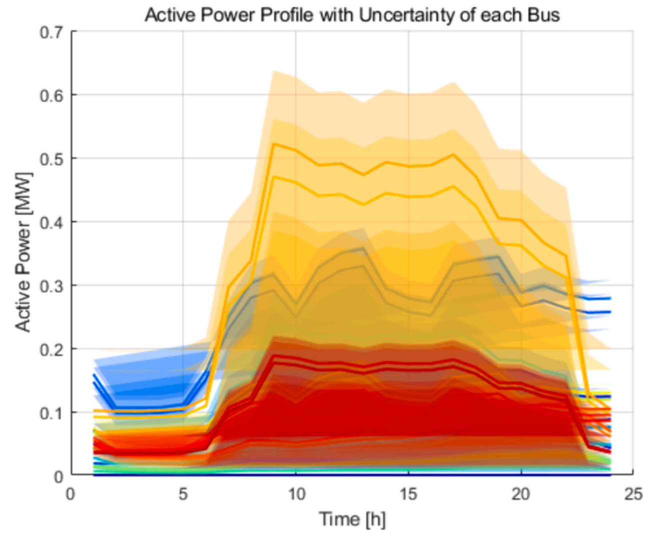


**Fig. 10.** Active power profile with uncertainty of each bus.

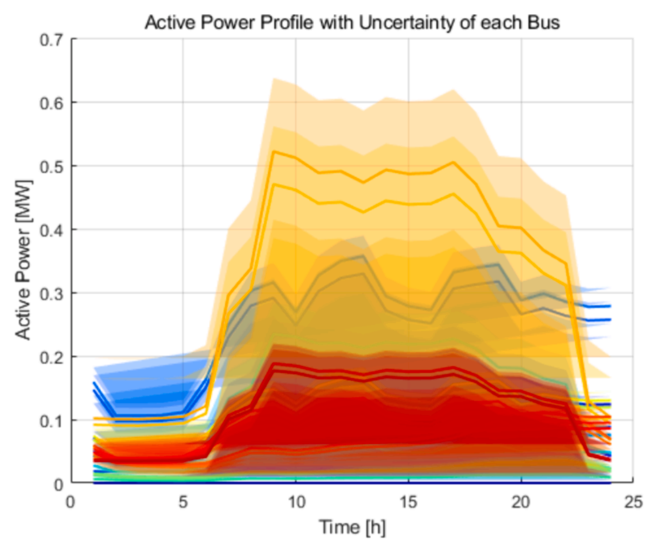


**Fig. 11.** Reactive power profile with uncertainty of each bus.

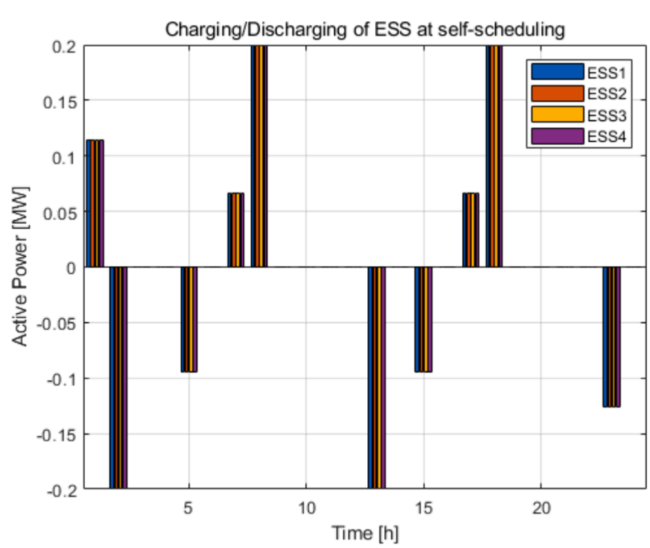


**Fig. 12.** ESS profile of ESS participants' self-scheduling.

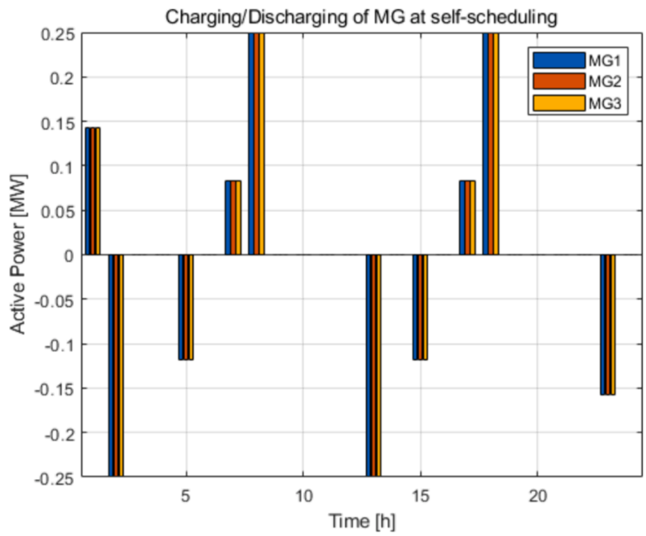


**Fig. 13.** ESS profile of MG participants' self-scheduling.

considered the same, they show similar behavior, except for the impact of uncertainty in the owned load or PV. All resources perform charging at low prices or discharging at high prices in response to the given price. Based on the output, the participation status in the LEM market is determined as a buyer or seller. Participants with only PV also have their status determined by their net load.

### *4.3. Local energy market clearing results*

In LEM, the matching result is represented as shown in Fig. 14. Most of the bidding occurred during hours of PV generation, except at 1:00 and 8:00, when participants on buses 17, 22, and 32, which have relatively large ESS, were successful. Clearing is performed by resources with high sell amounts and relatively low prices.

The clearing price of the LEM can be seen in Fig. 15. Since the prices of the bids were submitted by sellers and buyers randomly within the WEM price, it is observed that the clearing price is usually close to the buy price of the WEM, which is the upper limit, during times of high PV generation. This is because there are more buyers than sellers in the distribution network, as there are also resources that can act as buyers and general loads in the distribution network. Even so, the presence of LEMs proves to be more profitable than the prices of the seller and buyer WEMs.

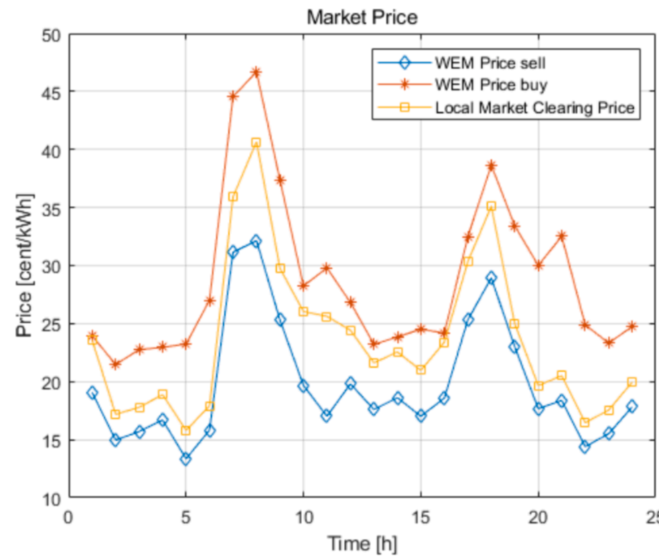


**Fig. 15.** LEM clearing price profile.

### *4.4. VPP and DSO coordination results*

The VPP bids on mismatched amounts from the LEM, potential additional buys and sells due to uncertainty, and adjusted output from the LFM. The results of the VPP bidding for this are shown in Fig. 16. The VPP's sells tend to occur during times of high PV generation. This is because participants use the VPP to internally supply mismatches in their buy from other resources in the LEM, even if they have a sell mismatch in the LEM.

The loss-sensitivity-based flexibility upward/downward prices for DERs is depicted in Fig. 17. Bus 13 with PV has a very low upward flexibility price and a very high downward flexibility price at 10:00, which indicates a relatively high load and a situation where the impact of losses is high and electricity purchases need to be curbed. Since PV does not provide upward flexibility, other resources with relatively low prices support it.

The results of LFM, where VPPs and DSOs perform coordination, are primarily notable in ESS. The change in self-scheduling of ESS in LFM is

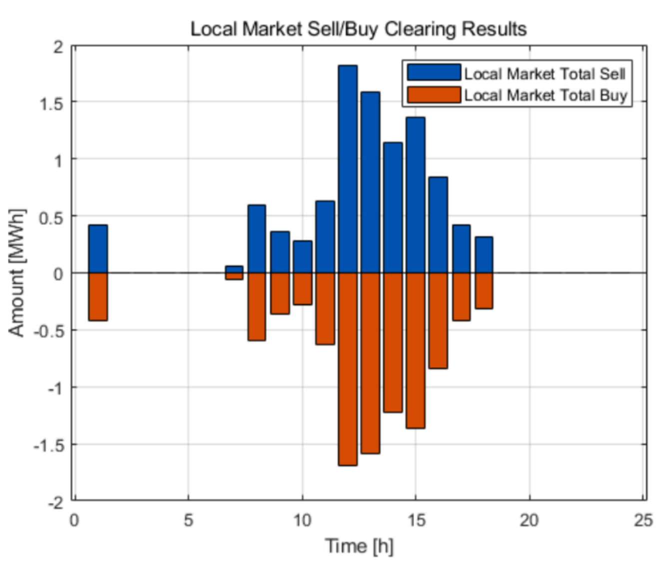


**Fig. 14.** LEM Clearing results.

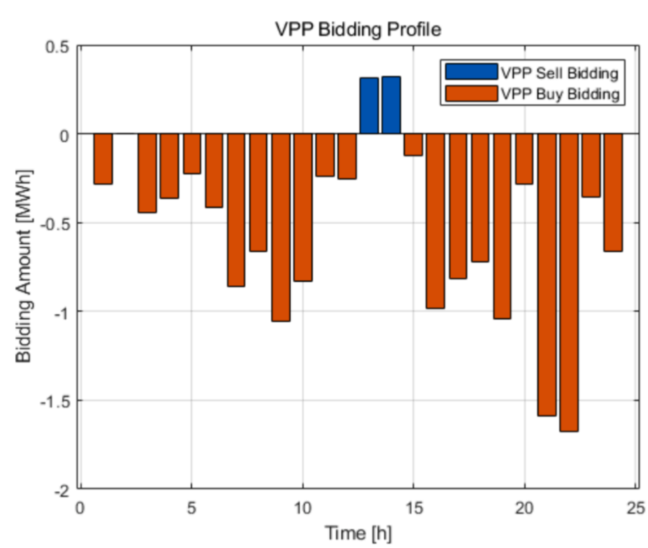


**Fig. 16.** Bidding profile of VPP.

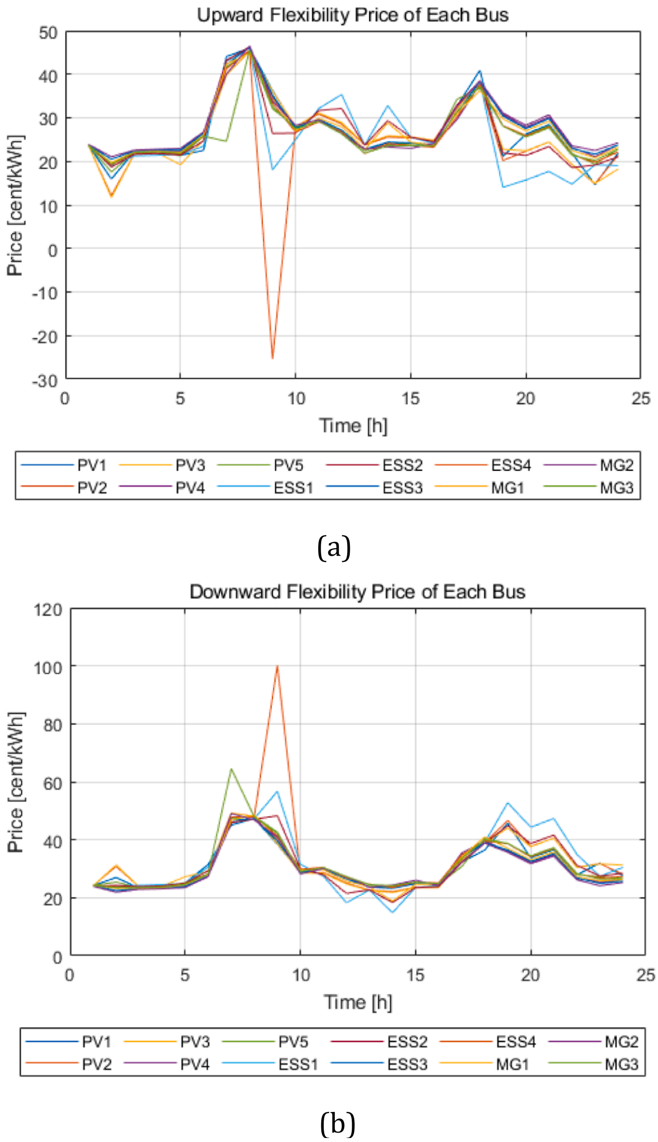


**Fig. 17.** (a)Upward/(b) downward LFM price for DERs.

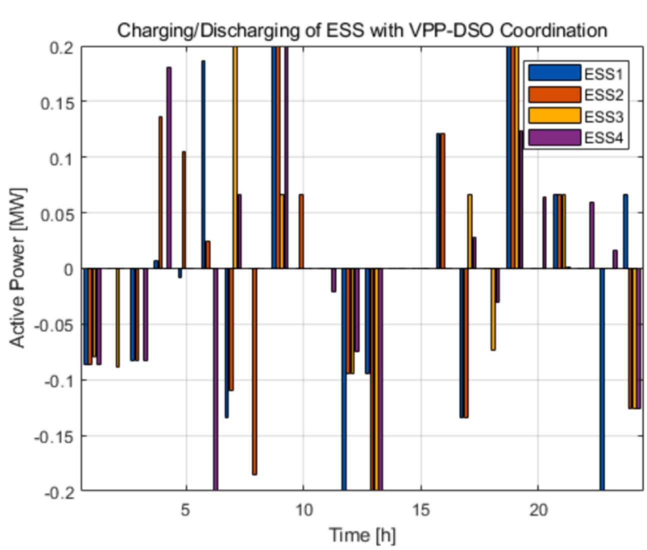


**Fig. 18.** ESS profile of ESS participants after LFM.

shown in Fig. 18. Compared to Fig. 12, the ESS provides flexibility to the distribution network by performing many actions. Flexibility supply from ESS 2 and 4 is very active, as there are several time points with relatively low flexibility prices. The overall behavior of ESSs varies depending on the conditions in the distribution network. At 1:00, it discharges in response to the price, but if the network is considered, it performs a charge at this time. Between 8:00 and 10:00, the voltage drops on some buses, so additional discharges are performed differently. During PV generation hours, charging is planned in self-scheduling, so it is performed without any other adjustment. Also, at 20:00, when the voltage drops significantly, the existing discharge plan is modified to perform more discharges. The additional behavior is due to the condition of maintaining the SOC at its initial value. On the other hand, for all MGs, there are situations where they do not supply flexibility for several reasons. First, MGs are located at a relatively greater distance from PV, which is the main source of problems in the distribution network. The flexibility compensation of MG resources is less than the best resources in most situations, as shown in Fig. 17. In addition, the provision of flexibility by MGs has the effect of adjusting the DSO's WEM purchases, but at the expense of a higher flexibility compensation cost. Therefore, the DSO prefers to obtain flexibility from VPP participants.

### *4.5. Voltage stability of distribution network*

In this section, the voltages for each case are compared. Fig. 19 shows the voltage in each case when the average value of the predicted PV and load is applied. In Case 1 and Case 2, because LFM is not considered, no voltage regulation occurs, resulting in many voltage violations. The difference between Case1 and Case2 is caused by the relative lack of resources available for Case2 to adjust because of the presence or absence of the self-scheduling step. On the other hand, in Case 3, more flexibility is provided in the generation period of PV and these adjustments have more impact on the VPP's profit, therefore the VPP makes more adjustments than in Case 4, regardless of the participants' decisions. Case 4 provides relatively reasonable flexibility to stabilize voltage in the distribution system. Because of the assurance of matching LEMs, it may have a somewhat lower range of adjustments, but it ensures the privacy of the participants more than the VPP's profit from adjustments.

The voltage stability analysis for each case based on the MCS-based scenario generation is summarized in Table 3. For each case, 100 scenarios were randomly generated within the prediction range. The voltage was analyzed based on AC power flow.

In Case 1 and Case 2, where network constraints are not considered, a very large number of voltage violations occur, which eventually lead to a decrease in profit for the participants as well as the VPP. The highest accuracy corresponds to Case 3. It is observed that the VPP operates all DERs in coordination with the stability of the distribution network, resulting in higher voltage satisfaction. In Case 4, where the framework proposed in this paper is applied, the accuracy is slightly lower than that of Case 3, but it is still very high. In particular, the difference between the two is almost zero, which shows that the operation of the proposed framework can also make a high contribution to the stability of the distribution network. On the other hand, due to the application of linearized network constraints, the exact voltage could not be calculated, resulting in a situation where the voltage satisfaction of all scenarios based on MCS was not satisfied.

The accuracy analysis of MIP-PF is summarized in Table 4. Most error indices show high accuracy. However, the MAPE is higher for line losses. This is because the linearized constraints cannot accurately consider line losses, which resulted in very large errors in certain situations. On the other hand, the voltage accuracy is very high, which proves that the application of MIP-PF is able to consider the correct voltage.

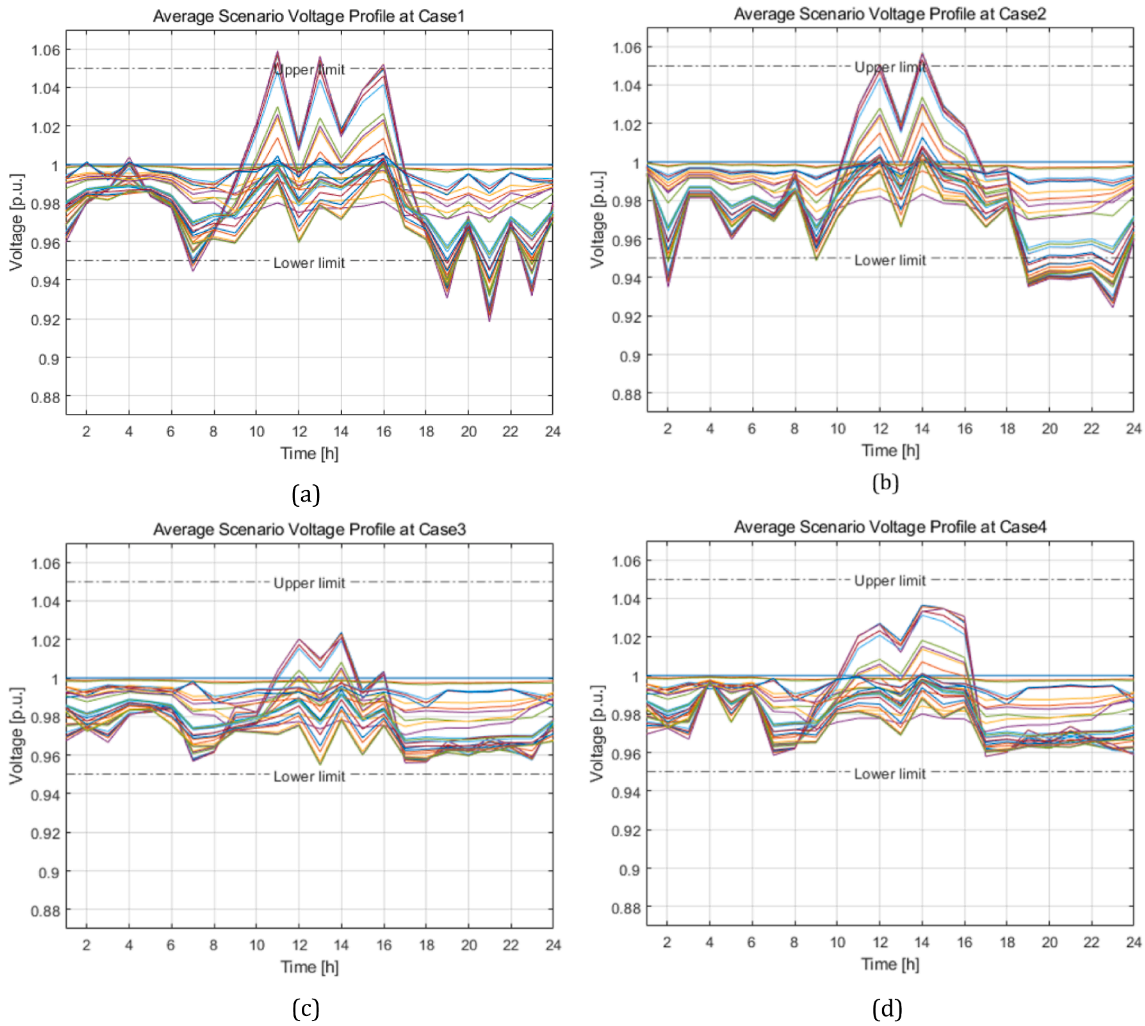


**Fig. 19.** Voltage profile of (a) case1, (b) case2, (c) case3 and (d) case4 at average scenario.

**Table 3**
Voltage satisfaction by case in MCS-based scenarios at IEEE 33 bus system.

| Case | Voltage satisfaction rate (%) |
|---|---|
| Case1 | 88.21 |
| Case2 | 89.02 |
| Case3 | 99.95 |
| Case4 | 99.86 |

**Table 4**
Power flow accuracy of MIP-PF method.

| Parameter | MSE | RMSE | MAE | MAPE |
|---|---|---|---|---|
| voltage | 5.667e-5 | 2.281e-4 | 0.0056 | 0.5703 % |
| active power loss | 7.092e-4 | 0.0266 | 0.0232 | 33.73 % |
| reactive Power loss | 5.775e-4 | 0.0240 | 0.208 | 41.76 % |

*4.6. Cost and benefit analysis*

The market profits and costs of the VPP for each case are summarized in Table 5. In Case 1, the VPP manages all resources, so the VPP's operations are the only profit generated. On the other hand, if the VPP allows participants to participate in the LEM, as in Case 2, the profits of

**Table 5**
Total cost/benefit of VPP and DERs by case at IEEE 33 bus system.

| Case | LEM Profit ($) | LFM Profit ($) | VPP Cost ($) |
|---|---|---|---|
| Case1 | – | – | −0.4559e3 |
| Case2 | 1.1103e3 | – | 0.1287e3 |
| Case3 | – | 4.3962e3 | 2.1450e3 |
| Case4 | 1.1103e3 | 4.8766e3 | 3.0279e3 |

the VPP are reduced. Although there is a benefit to the VPP participant from participating in LEM, compared to Case 1, the VPP has increased costs and no profit. In other words, in the absence of coordination between the VPP and DSO, it is considered a disincentive to give the participant market options. In contrast, Case 3 is beneficial due to the lowest cost of the VPP and the profit of the LFM. This shows that when VPPs interact with DSOs based on LFM, it is economical while satisfying network constraints. Finally, in Case 4, the high profit from participating in LFM and the profit from participating in LEM are generated in parallel. As a result, the VPP's profit is $1848.7, which is the difference between the operating cost and the LFM profit. Compared to Case 3, this is considered a small value compared to the difference of $2251.2, but if the participant is offered the LEM participation of $1110.3 in Case 4, the revenue earned by the VPP of $1140.9 is likely to be considered even smaller. Even considering that the actual VPP and participant profit sharing is considered in a ratio of about 6:4, the proposed framework shows that an appropriate distribution is achievable. Similarly, the profits that participants and VPPs consider are larger when compared to Case 1 and Case 2. As a result, giving VPP participants market options is a reasonable approach when the coordination of VPP-DSO is implemented on a market base.

### 4.7. Scalability

To verify the scalability of the proposed framework, the performance is validated on a larger system, the IEEE 69 bus system [60]. The IEEE 69 bus system is depicted in Fig. 20.

To demonstrate the performance of the proposed robust optimization, the same case as in the previous section was applied, and the voltage satisfaction for 100 random scenarios based on MCS on IEEE 69 bus system is summarized in Table 6. Overall, due to the basic configuration of the system, it does not lead to frequent voltage violations However, when the VPP-DSO coordination based on robust optimization is applied, the voltage satisfaction rate increases in both Case3 and Case4, indicating that the impact of uncertainty is considered. Furthermore, the results of Case3 and Case4 are not significantly different, showing that the proposed framework achieves respect for the network constraints while considering the profit.

Table 7 shows the profits and costs of VPP and DER by market. Similar results occur as discussed in the previous section. Case 1 shows the lowest cost of VPP. In Case 2, the cost of the VPP increases slightly, but the profits from participating in the LEM are higher, because the larger the network scale, the more LEM transactions possible. However, the VPP still exists as a cost and needs to be compensated by some LEM profits. On the other hand, in Case 3, the VPP's profit from LFM is $614.8, which is less than the total profit of $2885.4 compared to Case 2, even if the LEM profit is managed by the VPP. In Case 4, VPP's revenue from LFM is higher than in the previous cases, and the profit obtained by participants and VPP is higher. Although it is relatively low at $179.9, when considered together with the LEM profit, the total profit with participants increases to $3331.8. As the scale of the network increases, the percentage of profit that VPPs obtain from LFM decreases, but it is still higher than in the other cases.

## 5. Conclusion

In this paper, a hierarchical robust VPP-DSO coordination framework based on local markets is proposed. The applied local markets are classified into LEM and LFM. In LEM, members of the distribution network, such as VPP participants, MGs, and general loads, determine and submit their prices and demands based on self-scheduling. Then, clearing is performed based on a double-auction mechanism through the submitted information. The mismatches that are created after LEM clearing are aggregated by VPPs and bid on the WEM, while DSOs purchase surplus power from MGs and buy from WEM. The LFM market makes it possible for DSOs and VPPs to coordinate their operations to

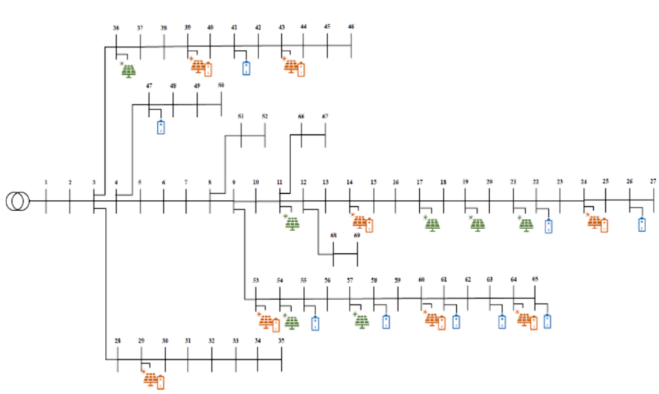

**Fig. 20.** Single line diagram of IEEE 69 bus system.

**Table 6**
Voltage satisfaction by case in MCS-based scenarios at IEEE 69 bus system.

| Case | Voltage satisfaction rate (%) |
|---|---|
| Case1 | 97.93 |
| Case2 | 98.04 |
| Case3 | 99.13 |
| Case4 | 99.12 |

**Table 7**
Total cost/benefit of VPP and DERs by case at IEEE 69 bus system.

| Case | LEM Profit ($) | LFM Profit ($) | VPP Cost ($) |
|---|---|---|---|
| Case1 | – | – | 0.2267e3 |
| Case2 | 3.1519e3 | – | 0.2665e3 |
| Case3 | – | 3.4062e03 | 2.7914e3 |
| Case4 | 3.1519e3 | 3.9191e03 | 3.7392e3 |

satisfy network constraints. Profits from LFM are compensated to VPPs through loss sensitivity-based flexibility prices. To avoid imbalance penalties in each market transaction, robust optimization was performed using a GP-based modeled uncertainty set of load and PV generation. The worst-case scenario considered in the LEM is the lowest renewable generation and the highest load. The worst-case scenario of VPP-DSO coordination is then considered in the form of interval optimization, which considers two cases for voltage stability of the distribution network. These are the minimum net load with the highest voltage and the maximum net load with the lowest voltage, respectively. The validation of the applied approach was verified in MATLAB environment with 100 scenarios based on MCS and IEEE benchmark system. The results are analyzed in respect of profit and voltage stability in four cases. In the case of no coordination between VPP and DSO, restricting LEM participation shows higher profit, but in the opposite case, the framework proposed in the paper performs better. The improved profit for the VPP and its participants from the perspective of total profit is freed from the complexity of profit allocation through methods and conflicts of interest. In addition, for voltage stability, the proposed approach results in voltage satisfaction at a level like that of a DSO directly managing VPP participants. In the scaled-up system, these patterns are similar, and the application of the framework proposed in this paper is better for voltage stability and for the profits of the VPP and its participants. On the other hand, this paper has the limitation that the self-scheduling is performed by the expected price during the LEM clearing process, and the selling and buying states are fixed. Also, the contribution of VPPs to the transmission system is limited by considering WEM as an energy market. Furthermore, since the uncertainty distribution is based on statistical methods, it is possible that the balance penalty is not accurately considered in real-time operation. Therefore, in future work, we will consider these aspects and implement a structure that can consider the interests of DER-VPP-DSO-TSO in the coordination problem formulation.

## CRediT authorship contribution statement

**Dongjun Han:** Conceptualization, Investigation, Methodology, Software, Validation, Visualization, Writing – review & editing. **Donghyun Koo:** Conceptualization, Methodology, Writing – review & editing. **Chankyu Shin:** Methodology, Validation, Writing – review & editing. **Dongjun Won:** Conceptualization, Methodology, Supervision, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

## Data availability

The data used in the manuscript is open source and the source is disclosed in the text.

## Acknowledgement

This work was supported by the Korea Institute of Energy Technology Evaluation and Planning (KETEP) and the Ministry of Trade, Industry & Energy (MOTIE) of the Republic of Korea (No. 20210501010010).

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. 2023R1A2C2005090).

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